

Osama Azab

AI Engineer and .NET Developer

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Summary

AI engineer and .NET developer. 10 years building .NET APIs and integrations for enterprise insurance platforms in Saudi Arabia, now adding AI to systems like these: agentic RAG, LLM agents, MCP servers and n8n automations. 1st place in Saudi Arabia at ACM Programming Jam X.

Experience

Software Engineer (Junior to Senior), Insurance technology company

Dec 2016 - Present, Riyadh

- Build .NET Web APIs, integration services and web portals for enterprise insurance platforms in Saudi Arabia: motor, medical and visit-visa insurance, from quotation to policy issuance, payment and renewal.
- Integrate with insurers, aggregators and Saudi government services (vehicle-accident data, national ID verification, social-insurance inquiries) over SOAP and REST. Integrated many payment gateways, including SADAD, and wrote their callback handling.
- Develop on and diagnose SQL Server behind high-traffic, business-critical systems: execution plans, index tuning, slow queries, blocking and deadlocks. Also Oracle and PostgreSQL.
- ASP.NET and .NET 6-8, RabbitMQ, Hangfire, Serilog, JWT, Azure DevOps. Top committer on the team (about 7,400 commits since 2019).

AI Engineering Co-op, Saudi health-insurance regulator

Jul 2025 - Dec 2025, Riyadh

- Built an agentic RAG assistant over internal policy documents: LangGraph, Docling, Qdrant and an open-source LLM, with NeMo Guardrails, Langfuse tracing and a Streamlit UI that shows sources and collects feedback.
- Built readmissions scoring and forecasting: LightGBM on about 650K claims, provider observed/expected scores and exponential-smoothing forecasts, on a Spark, Hive, Impala and Airflow pipeline with Power BI dashboards.
- Built n8n agent POCs with Ollama: a daily news digest, compliance logs to Jira tickets, and a multi-agent orchestrator.

Selected Projects

Open Security, AI code security scanner

github.com/gosamab/open-sec

- Desktop app (Rust, Tauri v2, SvelteKit) that runs a 5-stage Claude pipeline over a codebase with tree-sitter and exports findings as SARIF. Public, with releases.

Financer, accounting app with an MCP server

Own product

- .NET 8 modular monolith (DDD, CQRS, MediatR) on PostgreSQL with a double-entry ledger, a Vue 3 web app, and an MCP server so AI assistants can read reports and record transactions.

Makhtotah, Arabic manuscript OCR

Senior Graduation Project

- Deep learning plus hybrid binarization for Arabic historical documents, about 30% lower character error rate than the strongest commercial baseline on historical pages. Also fine-tuned Qwen2 on Arabic text lines.

Education

B.Sc. in Computer Science, Prince Sultan University (first honors)

Aug 2021 - Dec 2025, Riyadh

Skills

AI: LLM agents, RAG, LangChain, LangGraph, Anthropic and OpenAI APIs, MCP, n8n, Qdrant, Langfuse, NeMo Guardrails

Backend: C#, ASP.NET Core, .NET 8, EF Core, SOAP and REST, Python, FastAPI, Rust

Data: SQL Server, Oracle, PostgreSQL, Spark, Airflow, LightGBM, scikit-learn, Power BI

Frontend and tools: TypeScript, Next.js, Vue 3, SvelteKit, Git, Docker, Azure DevOps, GCP

Spoken: Arabic (native), English (fluent, IELTS 6.5)

Awards and Certifications

ACM Programming Jam X: 1st place, Saudi Arabia (national); **IEEE Xtreme 18.0:** 6th in KSA, 271st globally; **Programming Jam 9:** 5th place

Certifications: CS50 AI, CS50x (HarvardX), Oracle Database SQL Certified Associate, CompTIA ITF+, UI Certified Designer